CORNELL TECH | APPLIED MACHINE LEARNING

Human or AI?

Group 11: Final Project Report

Kaggle team: Group Number 11

Course GitHub PR: https://github.com/sekhari/human-or-ai-submissions/pull/13

Branch: `submission/group-number-11`

Submission status: Submitted to Kaggle; public score confirmed

Q0. Team information and model summary

Q0.1. Team members and main contributions.

Member	NetID	Contributions
Yilin Gao	yg695	Initial pipeline, model comparison and tuning (Q1); shared split and initial submissions.
Ansh Raj Bhargava	ab3396	Unlabeled-data experiments (Q3) and robustness improvements (Q5).
Yingzi Gu (Grace)	yg667	Learning curves and error analysis (Q2/Q4); final refinement, reproducibility and LaTeX integration, with AI assistance disclosed in Q6.

Q0.2. Final pipeline, from text to probability.

We concatenate word unigram/bigram TF-IDF and character 3–5-gram TF-IDF. A smoothed class log-count ratio, estimated only from labeled training documents, reweights each feature before fitting a linear SVM with $C = 0.5$. We selected this NB-weighted model using the fixed 2,000-row validation set, then retrained it on all 10,000 labeled examples for Kaggle. A sigmoid fitted to full-pipeline out-of-fold training margins converts scores to estimated AI probabilities. The final model does not use pseudo-labels, because the tested self-training candidate had lower validation AUROC.

Q0.3. Final scores.

Metric	Score
Selected model validation AUROC	0.977012
Selected model Kaggle public AUROC	0.85994
Selected model Kaggle private AUROC	Not yet available

The screenshot-confirmed earlier public score is 0.85049 for `submission_word_svm_c05.csv`. It is not the score of the selected NB-weighted model. No private score is inferred or fabricated.

All experiments use the committed 8,000/2,000 split and seed 42. No raw or case/whitespace-normalized exact cross-split duplicates were found; near duplicates were not exhaustively tested. Repeated selection can make validation scores optimistic.

Q1. Model comparison and selection

Q1.1. Preprocessing and text representation.

We use light preprocessing to preserve stylistic cues. The word vectorizer lowercases text and uses scikit-learn's default word-token pattern (tokens of at least two word characters), with unigrams and bigrams. Character features use lowercased character 3–5-grams and retain local punctuation and spacing patterns. Both vectorizers use minimum document frequency 2, sublinear term frequency, their default smoothed IDF and L2 normalization, with caps of 100,000 word and 150,000 character features. Each vocabulary and its IDF weights are learned exclusively from training data. Character features complement vocabulary with morphology and local formatting. The final model adds supervised NB feature weighting, defined in Q5; it does not apply the Unicode folding tested in the robustness experiment.

Q1.2. Comparison of three modeling approaches on the same split.

Model (word TF-IDF)	Train AUC	Val. AUC	Fit (s)
Linear SVM ($C = 1$)	1.000000	0.943355	2.16
Logistic regression ($C = 1$)	0.995359	0.934601	2.14
Complement NB ($\alpha = 1$)	0.984366	0.896508	1.91

Current-environment rerun. All three use word TF-IDF; timing includes feature extraction and classifier fitting, excluding evaluation. Historical teammate results are retained separately and differ slightly because the original environment was not pinned.

Q1.3. Hyperparameters for at least two approaches and selection protocol.

For logistic regression we searched $C \in \{0.1, 0.5, 1, 2, 5\}$; for the linear SVM, $C \in \{0.01, 0.1, 0.5, 1, 2\}$. All other settings and validation IDs were fixed. The best word-only values were $C = 5$ and $C = 0.5$, respectively. We selected by held-out AUROC without test labels. Full results are below; this is model selection on a reused holdout rather than an unbiased final test estimate.

Model	C	Train AUC	Val. AUC
Logistic Regression	0.1	0.959492	0.908819
Logistic Regression	0.5	0.986518	0.928824
Logistic Regression	1	0.995359	0.934601
Logistic Regression	2	0.999386	0.939157
Logistic Regression	5	0.999999	0.942735
Linear SVM	0.01	0.956640	0.905830
Linear SVM	0.1	0.995047	0.935096
Linear SVM	0.5	1.000000	0.943666
Linear SVM	1	1.000000	0.943355
Linear SVM	2	1.000000	0.942228

Q1.4. Final selection and subsequent changes.

The original selected model combined word and character TF-IDF with SVM $C = 0.5$, reaching 0.971228 in the current environment. We then compared six fixed refinement configurations (Q5). NB-weighted TF-IDF achieved 0.977012, a gain of 0.005784, with similar laptop-scale training cost, and was selected. Its label-dependent feature weights are refitted within every calibration fold. We retained the original supervised model for Q2's controlled learning-curve experiment, but reran all Q4 analyses on the final selected model. Q3 and Q5 negative results remain part of the experimental record rather than being overwritten.

Q2. Effect of training data size

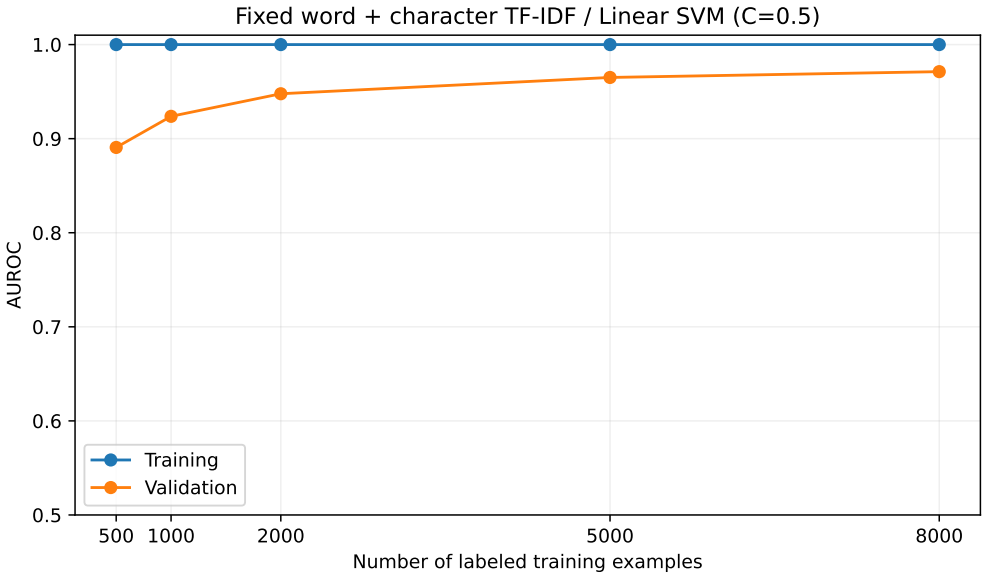
Q2.1. One fixed configuration at five training sizes.

We use the original word + character TF-IDF / SVM ($C = 0.5$) configuration throughout this experiment. This is the supervised reference configuration, not the later NB-weighted final model. Using seed 42, we independently shuffle each class pool and interleave them to obtain balanced, nested training subsets. The 8,000-row fit retains the original committed row order. Every size refits its own vocabulary and IDF weights, with identical hyperparameters and the same untouched validation IDs.

Training rows	Train AUC	Val. AUC	Fit (s)
500	1.000000	0.890672	0.63
1,000	1.000000	0.923734	1.28
2,000	1.000000	0.947738	2.62
5,000	1.000000	0.965055	6.26
8,000	1.000000	0.971228	9.77

Times include TF-IDF and SVM fitting, excluding scoring and probability calibration. Subset membership is recorded in `learning_curve_subset_ids.csv`.

Q2.2. Learning curve.



Q2.3. Overfitting, underfitting and the value of more labels.

Training AUROC is 1.0 at every size, while validation AUROC increases monotonically from 0.890672 to 0.971228. The initial gap of 0.109328 is consistent with overfitting in a high-dimensional representation, not an inability to fit the training observations. The gap falls to 0.028772 at 8,000 rows. There are diminishing returns: adding 3,000 examples from 2,000 to 5,000 improves validation AUROC by 0.017317; adding another 3,000 improves it by 0.006173. More labels from a similar distribution therefore appear useful, but this single seeded curve does not establish uncertainty or the gain beyond 8,000 examples. Additional in-distribution data may also fail to address unseen domains or adversarial editing.

Q3. Learning from unlabeled data

Q3.1. Method and leakage prevention.

Person B tested confidence-based self-training using the 3,000 unlabeled passages. The supervised word + character SVM predicts margins, transformed to probabilities with a logistic sigmoid fitted to five-fold out-of-fold training margins. Passages with estimated AI probability at least t or at most $1 - t$ receive pseudo-labels and are added to training. Thresholds $t \in \{0.99, 0.95, 0.90, 0.80, 0.50\}$ were compared, followed by up to three rounds at the best threshold. Vectorizer fitting, calibration, and pseudo-label generation exclude the fixed validation set. Validation labels select settings but do not train models. These exploratory calibration folds share a vocabulary fitted on the 8,000-row training partition; the revised final-refinement implementation instead refits the full pipeline per fold.

Q3.2. Supervised versus unlabeled-data results.

Method	Pseudo rows	Val. AUC	Time (s)
Supervised baseline	0	0.971122	64.85
Fit TF-IDF on training + unlabeled	0	0.971217	89.17
One round; confidence 0.99	1118	0.971113	75.49
One round; confidence 0.95	1849	0.971640	75.44
One round; confidence 0.9	2167	0.972072	75.65
One round; confidence 0.8	2479	0.972052	75.14
One round; confidence 0.5	3000	0.971796	75.32
self-training round 1; confidence 0.9	2167	0.972072	75.10
self-training round 2; confidence 0.9	2239	0.972232	87.93
self-training round 3; confidence 0.9	2255	0.972122	100.66

Person B's committed Kaggle-environment results, preserved without replacing them with another environment's numbers. Runtime includes feature preparation; later self-training rounds report cumulative runtime. The current-environment reference AUROC is 0.971228 versus historical 0.971122.

Q3.3. Did it help, and what limits the conclusion?

Self-training helped modestly: the best recorded two-round result was 0.972232 versus 0.971122 for supervised learning, a gain of 0.001110. At threshold 0.90, the first round added 2,167 pseudo-labeled passages; the second added 2,239. Lower thresholds admit more data but can introduce noise. In a training-only simulation, threshold 0.90 selected 2,022 of 3,000 held-back labeled examples with 1.88% error, compared with 9.6% when all were selected. This is an optimistic in-distribution diagnostic, not the error rate on genuinely unlabeled data. Threshold and round selection reuse the validation set, so the small gain is not independent confirmatory evidence.

Connection to final refinement

We also evaluated a reproducible two-round candidate with threshold 0.90 and a frozen sigmoid fitted from three-fold, full-pipeline out-of-fold training scores. Freezing avoids recalibrating on folds that contain pseudo-labels generated using all training targets. This revised procedure obtained 0.972186, above the current supervised baseline but below NB weighting. It differs from Person B's procedure and is reported as a separate experiment, not an exact reproduction. Its pseudo-label counts were 2,171 and 2,254. The final selected model therefore remains supervised.

Q4. Generalization and error analysis

Q4.1. Training versus validation performance of the final model.

The final NB-weighted model achieved training AUROC 0.999964 and validation AUROC 0.977012, a gap of 0.022952. This suggests some overfitting, while held-out ranking remains strong. High dimensionality, residual differences between training and validation passages, and repeated validation-based model selection are plausible contributors. No single cause is established. AUROC describes ranking, not accuracy at a fixed threshold. At the original SVM decision threshold of margin zero, the confusion matrix is:

True class	Predicted human	Predicted AI
Human	944	56
AI	106	894

There are 56 false positives and 106 false negatives: false-positive rate 5.6%, false-negative rate 10.6%, and accuracy 91.9%. The original unweighted model had 50 false positives and 127 false negatives, so improved ranking did not reduce both error types at the unchanged threshold.

Q4.2. Observable subgroups and differences.

We group passages by whitespace-separated word count, an observable quantity requiring no hidden domain information. The training length tertiles, 175 and 240 words, define short (≤ 175), medium (176–240), and long (> 240) groups. Ties stay together; boundaries were not chosen to maximize validation scores.

Group (words)	Count	Human / AI	Val. AUC	FP / FN
Short (76–175)	711	352 / 359	0.970135	29 / 38
Medium (176–240)	643	355 / 288	0.981289	22 / 27
Long (241–350)	646	293 / 353	0.983670	5 / 41

Short texts have the lowest AUROC (0.970135), while medium and long texts reach 0.981289 and 0.983670. Longer passages may provide more lexical and stylistic evidence, but this is an association, not a causal effect: topic, genre and class composition also differ. The long group still contains 41 false negatives at the fixed threshold. Group AUROC and threshold-dependent error counts should therefore be interpreted separately. Counts are included to show the amount of evidence behind each group.

Probability estimates and operating point

We refit the entire NB-weighted pipeline in each of three stratified training folds and fit a logistic sigmoid to its out-of-fold margins. The held-out validation set never enters calibration. The fitted transformation is

$$\widehat{P}(\text{AI} \mid s) = \text{sigmoid}(6.304971\, s + 0.117599).$$

We retain the SVM's decision rule $s > 0$, equivalent to $\widehat{P}(\text{AI}) > 0.529366$. This is not a probability threshold of 0.5. The increasing transformation preserves AUROC; its probabilities may still be imperfectly calibrated because fold models use fewer observations than the full fit. The final Kaggle pipeline separately recalibrates using out-of-fold scores from all 10,000 labeled examples.

Q4. Error examples and interpretation (continued)

Q4.3. Five false positives and five false negatives.

FP means human text predicted AI; FN means AI text predicted human. We inspect the five highest-margin FP and five lowest-margin FN, deliberately emphasizing confident failures. These examples are not a random or representative sample of all errors. Reasons below are hypotheses grounded in passage inspection, not proven feature attributions or a claim that dataset labels are wrong.

ID	Error	P(AI)	Observed features / possible failure
3671	FP	0.975171	Polished human biography: third-person facts and institutional names overlap with generated exposition.
3326	FP	0.972815	Human film review: formulaic praise and plot/cast/music structure resemble generated reviews.
3448	FP	0.962793	Human technical reference: repeated terminology and explanatory prose may receive AI-associated weights.
4587	FP	0.914680	Human forum question: structured hypothetical reasoning and repetition resemble generated explanations.
6567	FP	0.910239	Human historical summary: dates, named organizations, and broad concluding statements mimic AI summaries.
1524	FN	0.000124	AI abstract plus personal question: technical terminology and missing visual context imitate a copied forum discussion.
3062	FN	0.002571	AI personal narrative: greetings, uncertainty, awkward wording, and personal detail imitate human forum writing.
8986	FN	0.002896	AI recipe: ingredient lists and headings mimic human recipes; an incongruous personal ending is overlooked.
2209	FN	0.010841	AI recipe: quantities and ordinary procedural verbs resemble human instructions, with no obvious local anomaly.
3665	FN	0.021416	AI news-style passage repeats one claim extensively. Topic cues dominate; sublinear TF and normalization dampen repetition.

Q4.4. Recurring failure patterns and a targeted response.

Polished human reference prose and reviews can resemble AI exposition (3671, 3326, 3448, 6567), while AI imitates technical forum posts, personal narratives, and recipes (1524, 3062, 8986, 2209). Genre and presentation therefore provide ambiguous authorship cues. Example 3665 adds a distinct failure: a news-style passage repeats nearly the same sentence throughout. Local lexical features preserve news vocabulary, while sublinear term frequency and normalization dampen repetition; the representation does not explicitly encode duplicate-sentence frequency. A targeted next experiment would add sentence-duplication fraction and longest repeated-sentence run as training-derived features, then compare against the unchanged baseline on the fixed validation IDs. This proposal has not been tested here. The selected cases support hypotheses but do not estimate prevalence, and neither surface polish nor factual plausibility establishes authorship.

Q5. Open-ended improvements and insights

Q5.1. Hypotheses motivating improvements.

Person B hypothesized that Unicode folding and typo augmentation could reduce brittleness to edited AI text, as the assignment describes adversarially modified test passages. Word/character TF-IDF matches local forms, so visually similar characters or inserted invisible characters may change features. In a later refinement, we hypothesized that class-informed feature weighting could emphasize discriminative word and character patterns instead of treating TF-IDF dimensions equally before regularization. These are separate experiments: robustness is evaluated on both original and synthetically perturbed validation passages, whereas final selection uses the original fixed validation set. Neither hypothesis guarantees improvement on the hidden test distribution.

Q5.2. Changes and controlled comparisons.

For robustness, R0 is word-only SVM; R1 is the word + character baseline; R2 adds NFKC normalization, selected homoglyph folding, zero-width removal and whitespace normalization; R3 additionally augments both training classes with typos. All retain SVM $C = 0.5$ and the fixed split. The stress tests perturb only AI-labeled validation passages using typos, Cyrillic look-alikes, or inserted zero-width characters; they are synthetic conditions, not known test-set memberships.

For refinement, we specified six candidates: baseline, character weight 2, character-only features, logistic regression $C = 5$, NB weighting, and revised two-round self-training. With binary training-document occurrence counts n_{cj} for class c and feature j, NB weighting adapts Wang and Manning (2012) to TF-IDF:

$$p_{cj} = \frac{1 + n_{cj}}{\sum_k (1 + n_{ck})}, \qquad r_j = \log \frac{p_{1j}}{p_{0j}}, \qquad x'_j = x_j r_j.$$

The same word/character TF-IDF settings and SVM $C = 0.5$ are retained. Counts use training labels only and are refitted within calibration folds. Validation labels never define r_j.

Q5.3. Measured results.

Configuration	Original	Typos	Homoglyph	Zero-width	Fit (s)
R0	0.9433	0.9404	0.9027	0.9185	5.12
R1	0.9711	0.9696	0.9578	0.9600	28.53
R2	0.9597	0.9576	0.9597	0.9597	28.94
R3	0.9590	0.9573	0.9590	0.9590	53.96

Historical Person B run. All score columns are validation AUROC; original means unmodified validation. Runtime includes feature extraction and fitting.

Candidate	Train AUC	Val. AUC	Fit (s)
Original word + character SVM	1.000000	0.971228	9.80
Character block weighted 2x	1.000000	0.971349	9.78
Character-only SVM	0.999978	0.969718	8.27
Word + character logistic regression	1.000000	0.969331	12.05
NB-weighted word + character SVM	0.999964	0.977012	10.39
Self-training, two rounds (revised)	1.000000	0.972186	41.14

Current-environment refinement. Time includes fitting and, for self-training, its internal calibration; final output-probability calibration is separate. Test data were not accessed by the candidate-selection script.

Q5.4. Support for the hypotheses, limitations and next steps.

Unicode normalization reduced original-validation AUROC from 0.971122 to 0.959700, and typo augmentation did not recover it (0.959011). Homoglyph robustness improved slightly, but the reported paired interval included zero. We therefore did not adopt normalization merely because it seemed intuitively robust. NB weighting improved original-validation AUROC from 0.971228 to 0.977012. A paired 1,000-resample bootstrap gives a descriptive 95% interval of [0.001478,0.010103] for its AUROC gain. This interval is conditional on the selected candidate and is not adjusted for repeated model selection. Thus the result supports further testing but does not prove improvement on unseen generators or adversarial text. All six candidates and negative results remain recorded. The submitted pipeline scored 0.85994 publicly versus 0.85049 for the earlier word-only submission; this is not an isolated test of NB weighting. Private performance remains unknown.

Q6. Code, reproducibility and AI assistance

Q6.1. PR, entrypoints, dependencies and final predictions.

PR: https://github.com/sekhari/human-or-ai-submissions/pull/13

Run from `submissions/group-number-11/`, with data in `data/competition/`. Python 3.14.2; pinned dependencies; seed 42.

```
python -m pip install -r requirements-final.txt
python run_final_refinement.py --data_dir data/competition
python evaluate_final_model.py --data_dir data/competition
python generate_final_submission.py --data_dir data/competition
python verify_final_results.py --data_dir data/competition
python build_report.py
```

The final prediction file is `artifacts/submission.csv`. See README for Q1–Q5 commands and JSON manifests for hashes. Raw data, passages, caches and weights are excluded from Git. Compile `report/main.tex` using Tectonic, or upload the standalone ZIP to Overleaf and use pdfLaTeX. After Kaggle scoring, update `report/status.json`, regenerate tables and recompile. Private score remains unavailable until revealed.

Q6.2. Development history and lessons.

- 8b44412 (Q3/Q5): self-training helped slightly; normalization could hurt original-validation performance.
- c8e5c1f (Q2/Q4): more labels narrowed the gap; error inspection exposed ambiguous genre cues.
- e999954 (final refinement) adds NB weighting, six-candidate comparison and updated Q4; the following commits integrate LaTeX. All candidates and selection limitations are retained.

Q6.3. AI assistance and verification.

Codex assisted Grace with code, execution, debugging, passage inspection, hypotheses and LaTeX drafting. Automated checks validate split membership, metrics, calibration, IDs and submission hashes; the rendered PDF was visually checked. These are not claims of human review. Each member should review the report, understand the approach and confirm their own tool disclosure before submission. Results are measured, failure explanations are hypotheses, and unknown Kaggle scores remain unfilled. No external LLM API predicted test labels and no hidden labels were retrieved.

Post-submission audit and next candidate

Status. Q0–Q4 describe the submitted NB model (public AUROC 0.85994). This appendix records a subsequent independent review and an unsubmitted candidate; its Kaggle score is unknown.

What the review found

The high fixed-validation score did not establish generalization. Among the supplied labels, no human passage contains double newlines, while 3,301 of 5,000 AI passages do. Many large NB coefficients involve line breaks or template markers. Applying Unicode/quote/whitespace normalization to both validation classes reduces NB AUROC from 0.977012 to 0.917921; also removing punctuation gives 0.888169. These tests demonstrate sensitivity, not the exact cause of hidden-test errors.

An independent three-fold classifier distinguishing training text from test text achieves AUROC 0.730335, confirming observable distribution differences without test labels. Submission IDs and ranks were verified; the increasing sigmoid preserves AUROC. No exact cross-split text duplicates were found. Removing three validation examples with training-neighbor similarity at least 0.9 leaves AUROC 0.976943, insufficient to explain the public-score gap.

Independent generalization diagnostics

Using only the official 8,000 training texts, unigram TF-IDF, 48-dimensional SVD and K-means form 12 label-free text clusters. Three-fold GroupKFold holds complete clusters out. These are inferred text groups, not known hidden domains. An ordinary stratified three-fold control uses a similar training size. Neither diagnostic changes the official 2,000-row validation IDs.

Model	Fixed val.	Random CV	Cluster CV
Submitted NB model	0.977012	0.969897	0.903318
Style-only model	0.958415	0.955965	0.895069
75:25 NB/style blend	0.988136	0.984046	0.945723

CV values are unweighted means of three fold AUROCs. These are exploratory diagnostics; grouped folds differ in size and composition.

New model and evidence limits

The content-light style model uses normalized text, function-word proportions, lexical diversity, word/sentence-length distributions, punctuation/character statistics and sentence repetition. It is a histogram gradient-boosting classifier (220 iterations, learning rate 0.06, 15 leaves, minimum leaf size 30, L2 penalty 10). Each component margin is divided by its own training-score standard deviation; the selected blend uses 75% NB and 25% style. We compared four components and three fixed blends. Among candidates retaining the old fixed-validation AUROC, this blend has the best mean cluster-CV AUROC and improves every cluster fold. Selection is exploratory and reuses validation results; it is not preregistered confirmation.

The blend reaches fixed-validation AUROC 0.988136, with 26 false positives and 72 false negatives at margin zero. Its normalized-text stress AUROC is 0.964515; residual sensitivity remains. It is refitted on all 10,000 supplied labels, with full-pipeline three-fold calibration and no external labels. The separate file is `artifacts/submission_robust_blend.csv`. See `PROJECT_REVIEW.md` for findings and reproduction commands. Confirm the new submission before replacing final-model Q4.